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A PROJECT REVIEW ON

Experimental study on usage of recycled aggregates in concrete in building construction

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PROJECT GUIDE:B V HARIPRASAD SIR

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Abstract:

This project mainly discus about the concrete production by mixing recycle aggregate to get the required strength for different applications by replacing the natural coarse aggregate. The Concrete which is made by replacing the recycled coarse aggregate strength can be determined by few physical and mechanical characteristics. This project likewise talk about the examination of reuse aggregate with constant concrete blend of M25 and water cement proportion, similar to two unique ages have been taken and the age difference between those ages are of 30 years and few samples were caste and this two types of reuse aggregate is compared with conventional cement. This project center around the utilization of reused aggregate in concrete, this outcomes are truly laboratory based and they are created by crushing concrete specimen (cubes and prisms) at lab, for that 21 cubes and 21 prisms are casted and checked for 28 days, in that which 9 cubes and 9 prisms are casted with one age of reused aggregate with various substitution of natural coarse aggregate like 10 percentage, 15 percentage, and 20 percentage respectively.

Introduction:

As we everyone knows that concrete is the main construction material across the world wide. As 2/3rd of the concrete consists of aggregates, the skeleton form of the concrete is formed by the presence of aggregates only. In the amount of aggregate is the major part consist of coarse aggregate, so there is a huge demand on coarse aggregate, it makes the whole world to see towards the requirement of the coarse aggregate. if we extract the natural coarse aggregate it cause to the environmental imbalance, so the whole world is looking towards the aggregates which is extracted from the solid construction waste of the old building and the roads. To manage the concrete waste the usage of recycle coarse aggregate is very important step towards the development and reservation of the environmental. The property of the recycled aggregate is the main problem for using it because they will not fill the desire requirement as to alternatively replace with that of naturally coarse aggregate.

Literature Review:

- **Pavan P S**:- A study on recycle concrete aggregate In this journal, the results are obtained by conducting tests on hardened concrete are discussed. The tests are explained. Compressive strength test, split tensile strength test done on concrete cubes and cylinders and left for 28days for curing .The following tests are done on concrete with recycled coursed aggregates using OPC and PSC.

- **Mr. Tushar**:- Use of Recycled Aggregate Concrete have taken a study on use of recycled aggregate concrete and concluded that Utilization of reused total up to 30% does not influence the practical necessities of the structure according to the discoveries of the test outcomes. Different tests directed on reused totals and results contrasted and characteristic totals are tasteful according to IS 2386. Because of utilization of reused total in development, vitality and cost of transportation of characteristic assets and uncovering is fundamentally spared. This thus legitimately decreases the effect of waste material on condition.

METHODOLOGY:

The main source of recycled coarse aggregate are from the concrete structures which have exhausted there limit of serviceability, that may be either from the old houses, roads, industrial building, old hospitals, theaters, government building, when the structure is attracted to heavy fire accident's, earth quakes, land sliding's etc. The availability of recycle aggregate are very hard to find in the demolition waste but extracting the required size of aggregate are very difficult, and this extracted aggregate will not have same properties as same as natural aggregate. Finding the aggregate of different ages is a very typical task, this project mainly focus on the collection of two unique ages of recycled aggregate, so two ages of aggregates have been collect and named it as AGE 2 and AGE 2

Objective

- To find the strength of the concrete by rearranging the virgin coarse aggregate with reuse coarse aggregate with various percentages like 10 percentage, 15 percentage and 20 percentage with two different ages
- This experiment can help in study and as well as comparison of various properties of the aggregate through flexural strength and compressive strength.

AGE-1

This AGE 1 aggregate is collected from our college premises, its age is of about 10 years, the size of this aggregate are not unique, so it is first separated to the required size of the project requirement with the help of sieve, the size of the recycled coarse aggregate has been maintained in between 10 mm to 16 mm almost 140 kg of recycled aggregate have been collected and place nearby concrete lab for the further experiments.

1- Recycle aggregate s are separated for this waste at college

2- Collected aggregates from the concrete waste and it is named as AGE 1(10 YEARS)

Age-2

The AGE 2 is collected from the ESI government hospital which is located at EERAGADA, the age of the building is of 40 years old, the aggregate of this age is also not unique, so sieve has been done to bring the aggregate to the required size for the project. Few experiments has been done to the further use.

ESI Hospital

Pictures of ESI hospital while demolition and from where aggregates are collected.

Recycle aggregate after collection and sieved to a required sizes.

EXPERIMENTAL PROGRAMME:

COLLECTION OF MATERIALS

TESTING OF MATERIALS

MIX DESIGN CALCULATIONS
MIXING AT SPECIFIED PROPORTION

MOULDING

DEMOULDING AFTER 30 HOURS

CURING FOR 7 DAYS AND 28 DAYS

TESTING OF SPECIMEN

RESULT ANALYSIS & CONCLUSION

Three types of mix:

- Nominal mix – M5,M7.5, M10, M15
- Standard mix – M20
- Design mix – M25 , M30, M35, M40….M60

Design Mix M25 =1:2.1:3.2

Replacement percentage	Cement	Sand	Coarse aggregate	Recycled aggregate
10	2.4kg	5.07kg	7.62kg	0.752gms
15	2.4kg	5.07kg	7.62kg	1.143kgs
20	2.4kg	5.07kg	7.62kg	1.524kgs

Practical work:

1st stage

2nd stage

Final stage of cube

TESTS ARE USED

- Fineness test
- Consistency test
- Soundness of cement
- Compressive strength

Experimental Procedures:

- **MATERIALS USED:**
    1. Cement
    2. Fine aggregate(less than 4.75 mm)
    3. Coarse aggregate(of 12 mm)
    4. water

CEMENT:

It is the one of the important binding material used for preparing the concrete. It is made by mixing calcareous (lime) and argillaceous (clay) material in 2/3rd and 1/3rd proportion respectively. When water is added with cement a reaction(chemical) takes place, the nature of that reaction is an exothermic, a certain amount of heat is produced is known as heat of hydration. The better Portland cement exist in the following chemical proportions and ingredients like Alumina (6%), Silica (20%), Lime (63%, Iron oxide (3%), Magnesium oxide(2%). From the above chemical ingredient like Iron oxide, Alumina, Silica are called as Acidic materials whereas Magnesium and Lime are called as Alkaline materials a cert PPC 53 grade of cement has been used for this experiments this ordinary Portland cement of 53 grade has been conformed to IS 1226:1987. The tests conducted on cement and there results are given below.

Fine aggregate:

The sand which is available locally used for this project, and it is of wet aggregate, so for getting desire mix the dry sand quantity has been multiples by 1.35 time. Fine aggregate of size usually less than 4.75 mm.

COARSE AGGREGATE:

The aggregate of size 12 mm is used for this experiment on the basis of size of the cube and the prism. Coarse aggregate of usually more than 4.75 mm.

RECYCLED AGGREGATE:

The two ages of recycled aggregate has been collected and partial distribution has been done for the required size in between 16 to 10 mm.

SIEVE ANALYSIS OF RECYCLED COARSE AGGREGATE:

By allowing the sample downward from a series of standard sieves, each sieve will have some retained aggregates, the aggregates are separated into many groups, and each sieve will contains the aggregates in a particular size range.

Fineness test:

Fineness of the cement is known with dry sieving as per IS: 4031 (Part 1) – 1996. The fineness of cement is measured by sieving . The proportion of cement particle grain sizes of which is larger than the required mesh size is then determined.

CONSISTENCY TEST:

To find the amount of water used to make a cement paste of normal acceptable (consistency) as refer to Indian Standard: 4031 (Part 4) - 1988. The standard consistency of the paste(cement) can be done to allow the vicat plunger and make allow to penetrate up to a point of 0.5 to 0.7 cm from lower end of vicat mould and this has to be down with in the 3 to 5 minute after mixing cement paste .Generally the normal consistency of the cement glue paste has to be in between 25 to 30 percentage.

SOUNDNESS OF CEMENT:

To find the cement soundness by Le Chatelier method as per IS: 4031 (Part 3) - 1988.

After 28 days curing of recycled aggregate concrete

Compresiive strength of concrete

Experimental values of compressive strength at 28 days

S NO	Percentage taken	No of days	Average compressive strength at 28N/MM2
1	10%	28 days	38.0 N/MM2
2	15%	28 days	41.0 N/MM2
3	20%	28 days	36.5 N/MM2

RESULT AND DISCUSSION:

- The results which were obtained by conducting tests on the cubes and prism made with different percentage replacement of natural coarse aggregate with recycled coarse aggregate has shown that

1. The full replacement of recycle aggregate are showing the almost same result as of specimen casted with traditional concrete at 28 days curing
2. The replacement of natural coarse aggregate by adding 10 % of recycled aggregate are reaching the strength of the mix
3. According to the test results recycle aggregate will give the sufficient results and meet the standard strength of the specimen which were prepared with traditional aggregates
4. Mixing the natural coarse aggregate with recycle coarse aggregate with 15 %

Conclusion:

- Various tests conducted on RCA are compared with Indian code and the results are satisfactory and hence these can be used as aggregates
- Due to use of RCA in construction energy, cost of transportation are saved.
- Up to 50% replacing of RCA we get satisfactory strength
- Production cost decreases remarkably
- Due to lack of treatment of RCA adequate strength is not archived but by applying some treatment processes we can further improve the strength of the RCA
- Tensile test shows concrete has good tensile strength when replaced 50%
- Water absorption of RCA is high when compared with conventional aggregate.

References

- Concrete technology theory and practice for designing concrete mix-MS SHETTY
- IS456-2000 " code book for plain cement concrete-for minimum cement content in design mix and water cement ratio
- IS 10262-2009 "code book is used for recycled aggregates mix design

THANK YOU